

Table 1: Diastolic blood-pressure mean absolute error (mmHg) for every model across every dataset and protocol. CalFree, AAMI and the five external sets are calibration-free; CalBased shares 100% of its subjects with training and is an upper bound rather than a deployable figure. k is the number of per-subject calibration anchors. The final row gives the error of predicting each subject’s own mean, which most external columns barely improve on.

Model	Complexity	n	CalFree			CalBased	AAMI	MIMIC-BP	BCG	Sensors	UCI2	PPG-BP	Leans on
			$k=0$	$k=5$	$k=20$								
<i>Feature models (LightGBM)</i>													
GBM all features	50,400 lv	83	6.90	5.15	4.82	6.16	10.81	14.65	7.49	6.73	6.59	7.29	10th-percentile pulse height (how low the trough sits); pulse-shape asymmetry (lopsidedness of the wave)
GBM + demographics	50,400 lv	85	7.13	4.94	4.59	4.59	11.63	16.76	6.68	7.16	6.95	6.82	age; body mass index
GBM deep + demo	190,500 lv	85	7.32	4.97	4.48	3.80	11.00	17.41	6.66	6.89	6.86	6.95	age; body mass index
GBM no rate shortcut	50,400 lv	72	7.21	5.24	4.89	6.29	11.00	14.60	7.81	6.57	6.23	7.42	10th-percentile pulse height (how low the trough sits); pulse-shape asymmetry (lopsidedness of the wave)
GBM arrival time only	50,400 lv	6	8.47	6.33	6.04	8.23	16.11	11.09	6.73	6.20	5.94	8.95	pulse arrival time to the pulse peak; strength of ECG-PPG alignment
GBM single tree	32 lv	83	7.98	5.87	5.52	7.85	12.85	13.13	6.47	8.04	7.22	6.59	10th-percentile pulse height (how low the trough sits); timing of the reflected-wave shoulder
<i>Deep networks (raw waveform)</i>													
lenet1d	658,786 par	raw	7.45	5.32	5.07	6.67	11.87	–	–	–	–	–	learned from raw waveform
inception1d	472,450 par	raw	7.28	5.09	4.92	3.84	10.43	–	–	–	–	–	learned from raw waveform
xresnet1d50	886,850 par	raw	7.65	5.01	4.74	3.86	10.95	–	–	–	–	–	learned from raw waveform
xresnet1d101	1,805,122 par	raw	7.43	5.22	4.93	5.10	11.60	–	–	–	–	–	learned from raw waveform
transformer	107,010 par	raw	7.73	5.67	5.37	7.37	12.38	–	–	–	–	–	learned from raw waveform
Predict subject mean	0	–	–	–	–	4.98	4.45	4.66	2.21	3.93	1.08	0.00	–